

SOLIDWORKS API (.NET VBA C#) — Reference (129/337): IGtol (part 2 of 3)

SOLIDWORKS API Documentation (converted from CHM)

IGtol (part 2 of 3) (Type)

SOLIDWORKS API Help

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GetSymArcs Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymArcs Method (IGtol)

Gets an array that defines all arcs in the symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymArcs( _  
    ByVal SymIdx As System.Short _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Object  
  
value = instance.GetSymArcs(SymIdx)
```

C#

```
System.object GetSymArcs(  
    System.short SymIdx  
)
```

C++/CLI

```
System.Object^ GetSymArcs(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx

Indexed position of the symbol

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetSymArcs](#).

Remarks

Each arc in the symbol is defined by three points: center, arc start, and arc end.

Format of return information is the following array of doubles:

For the *i*th arc:

retval[9 * *i* + 0] = X-coordinate of center point

retval[9 * *i* + 1] = Y-coordinate of center point

retval[9 * *i* + 2] = Z-coordinate of center point

retval[9 * *i* + 3] = X-coordinate of arc start point

retval[9 * *i* + 4] = Y-coordinate of arc start point

retval[9 * *i* + 5] = Z-coordinate of arc start point

retval[9 * *i* + 6] = X-coordinate of arc end point

retval[9 * *i* + 7] = Y-coordinate of arc end point

retval[9 * *i* + 8] = Z-coordinate of arc end point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

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GetSymCircles Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymCircles Method (IGtol)

Gets an array that defines all circles in the symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymCircles( _  
    ByVal SymIdx As System.Short _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Object  
  
value = instance.GetSymCircles(SymIdx)
```

C#

```
System.object GetSymCircles(  
    System.short SymIdx  
)
```

C++/CLI

```
System.Object^ GetSymCircles(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

Array (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetSymCircles](#).

Remarks

Each circle in the symbol is defined by its radius and center point.

Format of return information is the following array of doubles.

For the i th circle:

$retval[4 * i] = \text{radius}$

$retval[4 * i + 1] = \text{X-coordinate of center point}$

$retval[4 * i + 2] = \text{Y-coordinate of center point}$

$retval[4 * i + 3] = \text{Z-coordinate of center point}$

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

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GetSymDesc Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymDesc Method (IGtol)

Gets the specified symbol description.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymDesc( _  
    ByVal SymIdx As System.Short _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.String  
  
value = instance.GetSymDesc(SymIdx)
```

C#

```
System.string GetSymDesc(  
    System.short SymIdx  
)
```

C++/CLI

```
System.String^ GetSymDesc(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

Symbol description

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetSymDesc](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

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GetSymEdgeCounts Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymEdgeCounts Method (IGtol)

Gets an array of the number of lines, arcs and circles in the specified symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymEdgeCounts( _  
    ByVal SymIdx As System.Short _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Object  
  
value = instance.GetSymEdgeCounts(SymIdx)
```

C#

```
System.object GetSymEdgeCounts(  
    System.short SymIdx  
)
```

C++/CLI

```
System.Object^ GetSymEdgeCounts(  
&    System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetSymEdgeCounts](#).

Remarks

Format of return information is the following array of short integers:

retval[0] = Line count

retval[1] = Arc count

retval[2] = Circle count

retval[3] = Text count

For more information, see [IGtol::GetSymText](#) and [IGtol::GetSymTextPoints](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

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GetSymLines Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymLines Method (IGtol)

Gets an array that defines all lines in the symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymLines( _  
    ByVal SymIdx As System.Short _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Object  
  
value = instance.GetSymLines(SymIdx)
```

C#

```
System.object GetSymLines(  
    System.short SymIdx  
)
```

C++/CLI

```
System.Object^ GetSymLines(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetSymLines](#).

Remarks

Each line in the symbol is defined by two points.

Format of return information is the following array of doubles:

For the i th line:

$retval[6 * i + 0]$ = X-coordinate of first point

$retval[6 * i + 1]$ = Y-coordinate of first point

$retval[6 * i + 2]$ = Z-coordinate of first point

$retval[6 * i + 3]$ = X-coordinate of second point

$retval[6 * i + 4]$ = Y-coordinate of second point

$retval[6 * i + 5]$ = Z-coordinate of second point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

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GetSymName Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymName Method (IGtol)

Gets the symbol name based on the specified index number that SOLIDWORKS uses to identify the symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymName( _  
    ByVal SymIdx As System.Short _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.String  
  
value = instance.GetSymName(SymIdx)
```

C#

```
System.string GetSymName(  
    System.short SymIdx  
)
```

C++/CLI

```
System.String^ GetSymName(  
    & System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
[Symbol index](#) number used by SOLIDWORKS

Return Value

Symbol name

 Visual Basic for Applications (VBA) Syntax

See [IGtol::GetSymName](#).

 See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

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GetSymText Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymText Method (IGtol)

Gets the symbol text.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetSymText( _  
    ByVal SymIdx As System.Short _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Object  
  
value = instance.GetSymText(SymIdx)
```

C#

```
System.object GetSymText(  
    System.short SymIdx  
)
```

C++/CLI

```
System.Object^ GetSymText(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
[Symbol index](#) number used by SOLIDWORKS

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetSymText](#).

Remarks

The format of the return data is an array of BSTRs. The size of the array is based on the number of text pieces in this GTol. This number can be determined using TextCount returned by [IGtol::GetSymEdgeCounts](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

SOLIDWORKS API Help

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GetSymTextPoints Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetSymTextPoints Method (IGtol)

Gets the text points for the specified symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymTextPoints( _  
    ByVal SymIdx As System.Short _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Object  
  
value = instance.GetSymTextPoints(SymIdx)
```

C#

```
System.object GetSymTextPoints(  
    System.short SymIdx  
)
```

C++/CLI

```
System.Object^ GetSymTextPoints(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx

Indexed position of the symbol

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetSymTextPoints](#).

Remarks

The size of the return array is based on the number of text pieces in this GTol, which you can determine using the return value from [IGtol::GetSymEdgeCounts](#).

The format of the returned data is an array of doubles:

```
retval[0] - X location of text 1
retval[1] - Y location of text 1
retval[2] - Z location of text 1

retval[3] - X location of text 2
retval[4] - Y location of text 2
retval[5] - Z location of text 2

retval[(n*3-3)] - X location of text n
retval[(n*3-2)] - Y location of text n
retval[(n*3-1)] - Z location of text n
```

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

SOLIDWORKS API Help

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GetText Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetText Method (IGtol)

Gets the specified text part of this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetText( _  
    ByVal WhichText As System.Integer _  
) As System.String
```

Visual Basic (Usage)

```
Dim instance As IGtol  
Dim WhichText As System.Integer  
Dim value As System.String  
  
value = instance.GetText(WhichText)
```

C#

```
System.string GetText(  
    System.int WhichText  
)
```

C++/CLI



```
System.String^ GetText(  
& System.int WhichText  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichText

Text part to get as defined in [swGTolTextParts_e](#)

Return Value

Text part

Visual Basic for Applications (VBA) Syntax

See [GTol::GetText](#).

Example

See [Set Text in Datum Tags and GTols](#) examples in [IGTol](#).

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::SetText Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTextAngleAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextAngleAtIndex Method (IGtol)

Gets the text angle for the specified piece of text in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextAngleAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.GetTextAngleAtIndex(Index)
```

C#

```
System.double GetTextAngleAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double GetTextAngleAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the piece of text

Return Value

Text angle for the specified piece of text in radians, measured CCW from the X axis

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetTextAngleAtIndex](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPoint Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTextAtIndex Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextAtIndex Method (IGtol)

Gets the specified text from this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextAtIndex(  
    ByVal Index As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.String  
  
value = instance.GetTextAtIndex(Index)
```

C#

```
System.string GetTextAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.String^ GetTextAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the text

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetTextAtIndex](#).

Example

[Get Text Items in GTol Frame \(VBA\)](#)
[Get Text Items in GTol Frame \(VB.NET\)](#)
[Get Text Items in GTol Frame \(C#\)](#)

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPoint Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::GetTextRefPositionAtIndex Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTextCount Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextCount Method (IGtol)

Gets the number of text items in the GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Integer  
  
value = instance.GetTextCount()
```

C#

```
System.int GetTextCount()
```

C++/CLI

```
System.int GetTextCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of text items

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTextCount](#).

Example

[Get Text Items in GTol Frame \(VBA\)](#)
[Get Text Items in GTol Frame \(VB.NET\)](#)
[Get Text Items in GTol Frame \(C#\)](#)

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)

[IGtol::GetTextPoint Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::GetTextRefPositionAtIndex Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTriangleAtIndex Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTextFont Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextFont Method (IGtol)

Gets the font for this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextFont() As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.String

value = instance.GetTextFont()
```

C#

```
System.string GetTextFont()
```

C++/CLI

```
System.String^ GetTextFont();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Name of the GTol font

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTextFont](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPoint Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::GetTextRefPositionAtIndex Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

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GetTextFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextFormat Method (IGtol)

Obsolete. Superseded by [IAnnotation::GetTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextFormat() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Object

value = instance.GetTextFormat()
```

C#

```
System.Object GetTextFormat()
```

C++/CLI

```
System.Object^ GetTextFormat();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTextFormat](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTextHeightAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextHeightAtIndex Method (IGtol)

Gets the text height for the specified piece of text in this Gtol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextHeightAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.GetTextHeightAtIndex(Index)
```

C#

```
System.double GetTextHeightAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double GetTextHeightAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the piece of text

Return Value

Text height for the specified piece of text in meters

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetTextHeightAtIndex](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPoint Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::GetTextRefPositionAtIndex Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

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GetTextInvertAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextInvertAtIndex Method (IGtol)

Gets the specified text item's invert flag.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextInvertAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Integer  
  
value = instance.GetTextInvertAtIndex(Index)
```

C#

```
System.int GetTextInvertAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.int GetTextInvertAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the piece of text

Return Value

Invert flag for the specified text item

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetTextInvertAtIndex](#).

Remarks

The invert flag specifies whether the text has been mirrored (reflected) about the X axis. Reflection is applied after text rotation.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextPoint Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::GetTextRefPositionAtIndex Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

SOLIDWORKS API Help

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GetTextPoint Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextPoint Method (IGtol)

Gets the Gtol text reference point, which is the upper-left corner of the bounding rectangle.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextPoint() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Object  
  
value = instance.GetTextPoint()
```

C#

```
System.Object GetTextPoint()
```

C++/CLI

```
System.Object^ GetTextPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTextPoint](#).

Remarks

Format of return information is the following array of doubles:

retval[0] = X-coordinate of text reference point

retval[1] = Y-coordinate of text reference point

retval[2] = Z-coordinate of text reference point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::GetTextRefPositionAtIndex Method](#)

GetTextPositionAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextPositionAtIndex Method (IGtol)

Gets the text item's offset relative to note's text point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextPositionAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetTextPositionAtIndex(Index)
```

C#

```
System.object GetTextPositionAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetTextPositionAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the piece of text

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTextPositionAtIndex](#).

Remarks

The return value is the following array of doubles:

[*textPtX*, *textPtY*, *textPtZ*]

where the text position values are offset values from the origin of this [IGtol](#) object.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPoint Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

GetTextRefPositionAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTextRefPositionAtIndex Method (IGtol)

Gets the reference position for the specified text item in this GTol (for example, upper-left, lower-left, center).

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextRefPositionAtIndex( _
    ByVal Index As System.Integer _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim Index As System.Integer
Dim value As System.Integer

value = instance.GetTextRefPositionAtIndex(Index)
```

C#

```
System.int GetTextRefPositionAtIndex(
    System.int Index
)
```

C++/CLI

```
System.int GetTextRefPositionAtIndex(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the piece of text

Return Value

Reference position of the specified text item as defined in [swTextPosition_e](#)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTextRefPositionAtIndex](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPoint Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::IGetTextPoint Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

GetTriangleAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTriangleAtIndex Method (IGtol)

Gets the triangle at the specified index.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetTriangleAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetTriangleAtIndex(Index)
```

C#

```
System.object GetTriangleAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetTriangleAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the piece of text

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTriangleAtIndex](#).

Remarks

The return value is the following array of doubles:

[*vertexPt1*[3], *vertexPt2*[3], *vertexPt3*[3], *isFilled*, *lineType*]

where:

vertexPt1[3] = First XYZ vertex point

vertexPt2[3] = Second XYZ vertex point

vertexPt3[3] = Third XYZ vertex point

isFilled = Boolean returned as a double and is True if the triangle is filled, false if not

lineType = Line type (see [swLineTypes_e](#))

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTriangleCount Method](#)
[IGtol::IGetTriangleAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTriangleCount Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetTriangleCount Method (IGtol)

Gets the number of triangles in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTriangleCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Integer  
  
value = instance.GetTriangleCount()
```

C#

```
System.int GetTriangleCount()
```

C++/CLI

```
System.int GetTriangleCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of triangles

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetTriangleCount](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTriangleAtIndex Method](#)
[IGtol::IGetTriangleAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUseDocTextFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetUseDocTextFormat Method (IGtol)

Obsolete. Superseded by [IAnnotation::GetUseDocTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetUseDocTextFormat() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Boolean  
  
value = instance.GetUseDocTextFormat()
```

C#

```
System.bool GetUseDocTextFormat()
```

C++/CLI

```
System.bool GetUseDocTextFormat();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetUseDocTextFormat](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetWitnessLine Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetWitnessLine Method (IGtol)

Gets the extension line that connects the leader of this geometric tolerance with the entity being dimensioned.

.NET Syntax

Visual Basic (Declaration)

```
Function GetWitnessLine() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Object

value = instance.GetWitnessLine()
```

C#

```
System.Object GetWitnessLine()
```

C++/CLI

```
System.Object^ GetWitnessLine();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

An array of six doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetWitnessLine](#).

Example

[Get GTol Witness Line \(VBA\)](#)
[Get GTol Witness Line \(VB.NET\)](#)
[Get GTol Witness Line \(C#\)](#)

Remarks

As you drag a geometric tolerance leader off an entity, SOLIDWORKS dynamically builds a witness or extension line that connects the leader to the entity.

The return value is the following array of six doubles:

[*startPt*[3], *endPt*[3]]

where:

- *startPt*[3] = x,y,z start point coordinates of the witness line
- *endPt*[3] = x,y,z end point coordinates of the witness line

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::IGetWitnessLine Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HasExtraLeader Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : HasExtraLeader Method (IGtol)

Gets whether the GTol has a bent leaderline or a straight leaderline.

.NET Syntax

Visual Basic (Declaration)

```
Function HasExtraLeader() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.HasExtraLeader()
```

C#

```
System.bool HasExtraLeader()
```

C++/CLI

```
System.bool HasExtraLeader();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if extra leader line exists, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::HasExtraLeader](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::IsAttached Method](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::GetLeaderCount Method](#)
[IGtol::GetLeaderInfo Method](#)
[IGtol::GetLeaderSide Method](#)
[IGtol::GetLineAtIndex Method](#)
[IGtol::IGetLeaderAtIndex2 Method](#)
[IGtol::IGetLeaderInfo Method](#)
[IGtol::IGetLineAtIndex Method](#)
[IGtol::SetLeader Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAnnotation Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetAnnotation Method (IGtol)

Gets the annotation for this specific GTol.

.NET Syntax

Visual Basic (Declaration)

Function IGetAnnotation() As Annotation

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As Annotation

value = instance.IGetAnnotation()
```

C#

```
Annotation IGetAnnotation()
```

C++/CLI

```
Annotation^ IGetAnnotation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IAnnotation](#) object

Visual Basic for Applications (VBA) Syntax

See [Gtol::IGetAnnotation](#).

Remarks

The IAnnotation object is a higher-level object that contains methods that are available to all types of annotations.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetAnnotation Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

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IGetArcAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetArcAtIndex Method (IGtol)

Gets information for the specified arc.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetArcAtIndex(  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim Index As System.Integer
Dim value As System.Double

value = instance.IGetArcAtIndex(Index)
```

C#

```
System.double IGetArcAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetArcAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the arc

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*lineType*, *startPt[3]*, *endPt[3]*, *centerPt[3]*, *rotationDir*]

where:

<i>lineType</i>	Line type
<i>startPt[3]</i>	XYZ arc start point
<i>endPt[3]</i>	XYZ arc end point
<i>centerPt[3]</i>	XYZ arc center point
<i>rotationDir</i>	Boolean returned as a double and represents the rotation direction, where CCW returns True and CW returns false

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetArcAtIndex Method](#)
[IGtol::GetArcCount Method](#)

SOLIDWORKS API Help

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IGetArrowHeadAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetArrowHeadAtIndex Method (IGtol)

Gets information on the specified arrow head in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetArrowHeadAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetArrowHeadAtIndex(Index)
```

C#

```
System.double IGetArrowHeadAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetArrowHeadAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the arrow head

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is an array of doubles in the following format:

[*arrowHeadPt*[3], *arrowHeadDir*[3], *arrowHeadWidth*, *arrowHeadHeight*, *arrowHeadStyle*]

where:

<i>arrowHeadPt</i> [3]	XYZ arrow head tip location
<i>arrowHeadDir</i> [3]	XYZ arrow head direction
<i>arrowHeadWidth</i>	Arrow head width where the width is measured along the arrow head direction
<i>arrowHeadHeight</i>	Arrow head height
<i>arrowHeadStyle</i>	Arrow head style as defined in swArrowStyle_e

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetArrowHeadAtIndex Method](#)
[IGtol::GetArrowHeadCount Method](#)
[IGtol::GetArrowHeadInfo Method](#)
[IGtol::IGetArrowHeadInfo Method](#)

SOLIDWORKS API Help

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IGetArrowHeadInfo Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetArrowHeadInfo Method (IGtol)

Gets an array of doubles that describe the geometry of the arrowhead on the far end of the leader line.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetArrowHeadInfo() As System.Double
```

Visual Basic (Usage)

```
Dim instance As IGtol  
Dim value As System.Double  
  
value = instance.IGetArrowHeadInfo()
```

C#



```
System.double IGetArrowHeadInfo()
```

C++/CLI

```
System.double IGetArrowHeadInfo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This information returned by this method is independent of whether this note actually has an arrowhead.

The format of the return information is the following array of doubles:

retval[0] = Arrow length (for example, leader into arrowhead)

retval[1] = Arrowhead length

retval[2] = Arrowhead width

retval[3] = Arrowhead style as defined in [swArrowStyle_e](#)

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetArrowHeadAtIndex Method](#)
[IGtol::GetArrowHeadCount Method](#)
[IGtol::GetArrowHeadInfo Method](#)
[IGtol::IGetArrowHeadAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAttachPos Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetAttachPos Method (IGtol)

Gets the attachment point of the GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAttachPos() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Double  
  
value = instance.IGetAttachPos()
```

C#

```
System.double IGetAttachPos()
```

C++/CLI

```
System.double IGetAttachPos();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method is meaningful only if this GTol is attached.

Format of return value is the following array of doubles:

retval[0] = X-coordinate of attachment point

retval[1] = Y-coordinate of attachment point

retval[2] = Z-coordinate of attachment point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetAttachPos Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFontInfo Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetFontInfo Method (IGtol)

Gets with the font information for this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFontInfo() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Double  
  
value = instance.IGetFontInfo()
```

C#

```
System.double IGetFontInfo()
```

C++/CLI

```
System.double IGetFontInfo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Format of return value is an array of doubles. Currently this consists only of a [line type index](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFontInfo Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFrameDiameterSymbols Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetFrameDiameterSymbols Method (IGtol)

Gets whether diameter symbols are displayed for the specified frame.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFrameDiameterSymbols( _  
    ByVal FrameNumber As System.Short _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNumber As System.Short  
Dim value As System.Boolean  
  
value = instance.IGetFrameDiameterSymbols(FrameNumber)
```

C#

```
System.bool IGetFrameDiameterSymbols(  
    System.short FrameNumber  
)
```

C++/CLI

```
System.bool IGetFrameDiameterSymbols(  
& System.short FrameNumber  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Frame number to examine (1 or 2)

Return Value

- in-process, unmanaged C++: Pointer to an array of Booleans (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value should be an array of two Booleans, with the following format:

retval[0] = True if the first tolerance value has a diameter symbol, false if it does not

retval[1] = True if the second tolerance value has a diameter symbol, false if it does not

Use this method in combination with [IGtol::GetFrameSymbols2](#) that is used to retrieve the frame symbols.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameSymbols2 Method](#)
[IGtol::GetFrameCount Method](#)
[IGtol::GetFrameValues Method](#)
[IGtol::IGetFrameSymbols2 Method](#)
[IGtol::IGetFrameValues Method](#)
[IGtol::SetFrameSymbols2 Method](#)
[IGtol::SetFrameValues Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFrameSymbols Method (IGtol)

[See Also](#)



Obsolete. Superseded by [IGtol::IGetFrameSymbols2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFrameSymbols( _  
    ByVal FrameNumber As System.Short _  
) As System.Short
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNumber As System.Short  
Dim value As System.Short  
  
value = instance.IGetFrameSymbols(FrameNumber)
```

C#

```
System.short IGetFrameSymbols(  
    System.short FrameNumber  
)
```

C++/CLI

```
System.short IGetFrameSymbols(  
& System.short FrameNumber  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Visual Basic for Applications (VBA) Syntax

See [Gtol::IGetFrameSymbols](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFrameSymbols2 Method (IGtol)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IGtol Interface](#) : IGetFrameSymbols2 Method (IGtol)

Obsolete. Superseded by [IGtol::GetFrameSymbols3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFrameSymbols2( _  
    ByVal FrameNumber As System.Short _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNumber As System.Short  
Dim value As System.String
```

```
value = instance.IGetFrameSymbols2(FrameNumber)
```

C#

```
System.string IGetFrameSymbols2(  
    System.short FrameNumber  
)
```

C++/CLI

```
System.String^ IGetFrameSymbols2(  
& System.short FrameNumber  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Frame number to examine (1 or 2)

Return Value

- in-process, unmanaged C++: Pointer to an array of strings (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return array is an array of six strings in the following format:

retval[0] = Geometric characteristic symbol
retval[1] = Material condition symbol for first tolerance value
retval[2] = Material condition symbol for second tolerance value
retval[3] = Material condition symbol for datum1
retval[4] = Material condition symbol for datum2
retval[5] = Material condition symbol for datum3

The character strings returned in the array correspond to symbols defined in **C:\ProgramData\SolidWorks\SolidWorks 20nn\langlenglish\gtol.sym**. The format of each string is <LibraryName-SymbolName> (for example, <GTOL-ANGULAR>, which is the angularity symbol from the ASME Geometric Tolerancing Symbols library).

Use this method with [IGtol::IGetFrameDiameterSymbols](#), which determines whether diameter symbols are displayed.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameCount Method](#)
[IGtol::GetFrameDiameterSymbols Method](#)
[IGtol::GetFrameSymbols2 Method](#)
[IGtol::GetFrameValues Method](#)
[IGtol::IGetFrameDiameterSymbols Method](#)
[IGtol::IGetFrameValues Method](#)
[IGtol::SetFrameSymbols2 Method](#)
[IGtol::SetFrameValues Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFrameValues Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetFrameValues Method (IGtol)

Gets an array that describes the text that appears in each of the fields of the specified GTol frame.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFrameValues( _
    ByVal FrameNumber As System.Short _
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim FrameNumber As System.Short
Dim value As System.String

value = instance.IGetFrameValues(FrameNumber)
```

C#

```
System.string IGetFrameValues(
    System.short FrameNumber
)
```

C++/CLI

```
System.String^ IGetFrameValues(
    & System.short FrameNumber
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Frame number to examine (1 or 2)

Return Value

- in-process, unmanaged C++: Pointer to an array of strings (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Format of return array of BSTRs is:

retval[0] = Tolerance 1

retval[1] = Tolerance 2

retval[2] = Datum 1

retval[3] = Datum 2

retval[4] = Datum 3

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameCount Method](#)
[IGtol::GetFrameDiameterSymbols Method](#)
[IGtol::GetFrameSymbols2 Method](#)
[IGtol::GetFrameValues Method](#)
[IGtol::IGetFrameDiameterSymbols Method](#)
[IGtol::IGetFrameSymbols2 Method](#)
[IGtol::SetFrameSymbols2 Method](#)
[IGtol::SetFrameValues Method](#)

SOLIDWORKS API Help

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IGetLeaderAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetLeaderAtIndex Method (IGtol)

Obsolete. Superseded by [IGtol::IGetLeaderAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLeaderAtIndex( _  
    ByVal Index As System.Integer, _  
    ByRef PointCount As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim PointCount As System.Integer  
Dim value As System.Double  
  
value = instance.IGetLeaderAtIndex(Index, PointCount)
```

C#

```
System.double IGetLeaderAtIndex(  
    System.int Index,  
    out System.int PointCount  
)
```

C++/CLI

```
System.double IGetLeaderAtIndex(  
& System.int Index,  
& [Out] System.int PointCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

PointCount

Visual Basic for Applications (VBA) Syntax

See [Gtol::IGetLeaderAtIndex](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetLeaderAtIndex2 Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetLeaderAtIndex2 Method (IGtol)

Gets information about the specified leader on this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLeaderAtIndex2( _  
    ByVal Index As System.Integer, _  
    ByRef PointCount As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim PointCount As System.Integer  
Dim value As System.Double
```

```
value = instance.IGetLeaderAtIndex2(Index, PointCount)
```

C#

```
System.double IGetLeaderAtIndex2(  
    System.int Index,  
    out System.int PointCount  
)
```

C++/CLI

```
System.double IGetLeaderAtIndex2(  
& System.int Index,  
& [Out] System.int PointCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the leader

PointCount

Number of (x,y,z) points returned in the array

Return Value

- in-process, unmanaged C++: Pointer to array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

A GTol can have a straight leader (1 segment) or a bent leader (2 segments), or it might have a perpendicular leader, which can have either 2 or 3 segments, depending on the situation. The returned array contains the point information to define the leader. It contains the x,y,z coordinates for each vertex in the leader. It can contain up to 12 doubles (4 sets of x,y,z values).

Because the return value is passed into this method, it should be dimensioned to 12 before this method is called to allow for the maximum number of items that can be returned (4 segments). Upon return from the API, the PointCount argument contains the actual number of points stored in the array. Multiply this number by 3 to determine the number of array items actually used.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::GetLeaderCount Method](#)
[IGtol::GetLeaderSide Method](#)
[IGtol::IGetLeaderInfo Method](#)
[IGtol::SetLeader Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

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IGetLeaderInfo Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetLeaderInfo Method (IGtol)

Gets information describing the layout of the GTol leader lines.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLeaderInfo( _  
    ByRef PointCount As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim PointCount As System.Integer
Dim value As System.Double

value = instance.IGetLeaderInfo(PointCount)
```

C#

```
System.double IGetLeaderInfo(
    out System.int PointCount
)
```

C++/CLI

```
System.double IGetLeaderInfo(
    & [Out] System.int PointCount
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointCount

Number of (x,y,z) points returned in the array

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [Gtol::IGetLeaderInfo](#).

Remarks

There can be 0, 1 or 2 lines with the leader line. If the GTol is not attached, then there are 0 lines. Otherwise, the GTol can have a straight leaderline (1 line) or a bent leaderline (2 lines). The caller must infer the number of leader lines based on [IGtol::IsAttached](#) and [IGtol::HasExtraLeader](#).

- `IGtol::IsAttached() == false` implies no leaderline; therefore, there are no leaderline points (`PointCount=0`).
- `IGtol::HasExtraLeader() == false` means that this is a straight leaderline; therefore, there is 1 line (`PointCount=2`).
- `IGtol::HasExtraLeader() == True` means that this is a bent leaderline; therefore, there are 2 lines (`PointCount=3`).

The format of return information is the following array of doubles:

`retval[0]` = X-coordinate of first leader point

`retval[1]` = Y-coordinate of first leader point

`retval[2]` = Z-coordinate of first leader point

`retval[3]` = X-coordinate of second leader point

`retval[4]` = Y-coordinate of second leader point

`retval[5]` = Z-coordinate of second leader point

`retval[6]` = X-coordinate of third leader point

`retval[7]` = Y-coordinate of third leader point

`retval[8]` = Z-coordinate of third leader point

Use [IGtol::GetLeaderCount](#) to see how many leaders are on the [IGtol](#) object.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::GetLeaderCount Method](#)
[IGtol::GetLeaderInfo Method](#)

IGetLineAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetLineAtIndex Method (IGtol)

Gets the start and end information for the specified line.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLineAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetLineAtIndex(Index)
```

C#

```
System.double IGetLineAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetLineAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the line

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*lineType*, *startPt*[3], *endPt*[3]]

where:

- *lineType* = the [line type](#)
- *startPt*[3] = the XYZ line start point
- *endPt*[3] = the XYZ line end point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLineAtIndex Method](#)
[IGtol::GetLineCount Method](#)

IGetNextGTOL Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetNextGTOL Method (IGtol)

Gets the next GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNextGTOL() As Gtol
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As Gtol  
  
value = instance.IGetNextGTOL()
```

C#

```
Gtol IGetNextGTOL()
```

C++/CLI

```
Gtol^ IGetNextGTOL();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the next [GTol](#)

Visual Basic for Applications (VBA) Syntax

See [Gtol::IGetNextGTOL](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetNextGTOL Method](#)
[IAnnotation::IGetSpecificAnnotation Method](#)
[IAnnotation::IGetSpecificAnnotation Method](#)
[IDrawingDoc::INewGtol Method](#)
[IDrawingDoc::NewGtol Method](#)
[IView::GetFirstGTOL Method](#)
[IView::IGetFirstGTOL Method](#)

SOLIDWORKS API Help

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IGetSymArcs Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetSymArcs Method (IGtol)

Gets an array that defines all arcs in the symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymArcs( _  
    ByVal SymIdx As System.Short _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Double
```

```
value = instance.IGetSymArcs(SymIdx)
```

C#

```
System.double IGetSymArcs(  
    System.short SymIdx  
)
```

C++/CLI

```
System.double IGetSymArcs(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Each arc in the symbol is defined by three points: center, arc start, and arc end.

Format of return information is the following array of doubles:

For the *i*th arc:

retval[9 * *i* + 0] = X-coordinate of center point

retval[9 * *i* + 1] = Y-coordinate of center point

retval[9 * *i* + 2] = Z-coordinate of center point

retval[9 * *i* + 3] = X-coordinate of arc start point

retval[9 * *i* + 4] = Y-coordinate of arc start point

retval[9 * *i* + 5] = Z-coordinate of arc start point

retval[9 * *i* + 6] = X-coordinate of arc end point

retval[9 * *i* + 7] = Y-coordinate of arc end point

retval[9 * *i* + 8] = Z-coordinate of arc end point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)

SOLIDWORKS API Help

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IGetSymCircles Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetSymCircles Method (IGtol)

Gets an array that defines all circles in the symbol.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetSymCircles( _  
    ByVal SymIdx As System.Short _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Double  
  
value = instance.IGetSymCircles(SymIdx)
```

C#

```
System.double IGetSymCircles(  
    System.short SymIdx  
)
```

C++/CLI

```
System.double IGetSymCircles(  
& System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Each circle in the symbol is defined by its radius and center point.

Format of return information is the following array of doubles.

For the *i*th circle:

retval[4 * *i*] = radius

retval[4 * *i* + 1] = X-coordinate of center point

retval[4 * *i* + 2] = Y-coordinate of center point

retval[4 * *i* + 3] = Z-coordinate of center point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetSymTextPoints Method](#)



Gets an array of the number of lines, arcs and circles in the specified symbol.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetSymEdgeCounts( _  
    ByVal SymIdx As System.Short _  
) As System.Short
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Short  
  
value = instance.IGetSymEdgeCounts(SymIdx)
```

C#

```
System.short IGetSymEdgeCounts(  
    System.short SymIdx  
)
```

C++/CLI

```
System.short IGetSymEdgeCounts(  
    & System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

- in-process, unmanaged C++: Pointer to an array of shorts (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Format of return information is the following array of short integers:

```
retval[0] = Line count  
retval[1] = Arc count  
retval[2] = Circle count  
retval[3] = Text count
```

For more information, see [IGtol::IGetSymText](#) and [IGtol::IGetSymTextPoints](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymLines Method](#)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetSymLines Method (IGtol)

Gets an array that defines all lines in the symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymLines( _  
    ByVal SymIdx As System.Short _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.Double  
  
value = instance.IGetSymLines(SymIdx)
```

C#

```
System.double IGetSymLines(  
    System.short SymIdx  
)
```

C++/CLI

```
System.double IGetSymLines(  
&    System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

Each line in the symbol is defined by two points.

Format of return information is the following array of doubles:

For the *i*th line:

retval[6 * *i* + 0] = X-coordinate of first point

retval[6 * *i* + 1] = Y-coordinate of first point

retval[6 * *i* + 2] = Z-coordinate of first point

retval[6 * *i* + 3] = X-coordinate of second point

retval[6 * *i* + 4] = Y-coordinate of second point

retval[6 * *i* + 5] = Z-coordinate of second point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)

[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymEdgeCounts Method](#)
[IGtol::IGetSymText Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

SOLIDWORKS API Help

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IGetSymText Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetSymText Method (IGtol)

Gets the symbol text.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymText( _  
    ByVal SymIdx As System.Short _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim SymIdx As System.Short  
Dim value As System.String  
  
value = instance.IGetSymText(SymIdx)
```

C#

```
System.string IGetSymText(  
    System.short SymIdx  
)
```

C++/CLI

```
System.String^ IGetSymText(  
&    System.short SymIdx  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx

[Symbol index](#) number used by SOLIDWORKS

Return Value

- in-process, unmanaged C++: Pointer to an array of strings. The size of the array is based on the number of text pieces in this GTol. This number can be determined using [TextCount](#) returned by [IGtol::IGetSymEdgeCounts](#).
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)
[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::GetSymTextPoints Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymTextPoints Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSymTextPoints Method (IGtol)

[See Also](#)



Gets the text points for the specified symbol.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetSymTextPoints( _
    ByVal SymIdx As System.Short _
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IGtol
Dim SymIdx As System.Short
Dim value As System.Double

value = instance.IGetSymTextPoints(SymIdx)
```

C#

```
System.double IGetSymTextPoints(
    System.short SymIdx
)
```

C++/CLI

```
System.double IGetSymTextPoints(
    & System.short SymIdx
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymIdx
Indexed position of the symbol

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The size of the return array is based on the number of text pieces in this GTol, which you can determine using the return value from [IGtol::IGetSymEdgeCounts](#).

The format of the returned data is an array of doubles:

retval[0] - X location of text 1

retval[1] - Y location of text 1

retval[2] - Z location of text 1

retval[3] - X location of text 2

retval[4] - Y location of text 2

retval[5] - Z location of text 2

retval[(n*3-3)] - X location of text n

retval[(n*3-2)] - Y location of text n

retval[(n*3-1)] - Z location of text n

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetSymArcs Method](#)
[IGtol::GetSymCircles Method](#)
[IGtol::GetSymDesc Method](#)
[IGtol::GetSymEdgeCounts Method](#)

[IGtol::GetSymLines Method](#)
[IGtol::GetSymName Method](#)
[IGtol::GetSymText Method](#)
[IGtol::IGetSymArcs Method](#)
[IGtol::IGetSymCircles Method](#)
[IGtol::IGetSymLines Method](#)
[IGtol::IGetSymText Method](#)

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IGetTextFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetTextFormat Method (IGtol)

Obsolete. Superseded by [IAnnotation::IGetTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTextFormat() As TextFormat
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As TextFormat  
  
value = instance.IGetTextFormat()
```

C#

```
TextFormat IGetTextFormat()
```

C++/CLI

```
TextFormat^ IGetTextFormat();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Gtol::IGetTextFormat](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

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IGetTextPoint Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetTextPoint Method (IGtol)

Gets the Gtol text reference point, which is the upper-left corner of the bounding rectangle.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTextPoint() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Double  
  
value = instance.IGetTextPoint()
```

C#

```
System.double IGetTextPoint()
```

C++/CLI

```
System.double IGetTextPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Format of return information is the following array of doubles:

retval[0] = X-coordinate of text reference point

retval[1] = Y-coordinate of text reference point

retval[2] = Z-coordinate of text reference point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTextAngleAtIndex Method](#)
[IGtol::GetTextAtIndex Method](#)
[IGtol::GetTextCount Method](#)
[IGtol::GetTextFont Method](#)
[IGtol::GetTextHeightAtIndex Method](#)
[IGtol::GetTextInvertAtIndex Method](#)
[IGtol::GetTextPositionAtIndex Method](#)
[IGtol::GetTextRefPositionAtIndex Method](#)
[IGtol::IGetTextPositionAtIndex Method](#)

SOLIDWORKS API Help

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IGetTextPositionAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetTextPositionAtIndex Method (IGtol)

Gets the reference position for the specified text item in this GTol (for example, upper-left, lower-left, center).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTextPositionAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetTextPositionAtIndex(Index)
```

C#

```
System.double IGetTextPositionAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetTextPositionAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the piece of text

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*textPtX*, *textPtY*, *textPtZ*]

where the text position values are offset values from the origin of this [IGTol](#) object.

See Also

[IGTol Interface](#)
[IGTol Members](#)
[IGTol::GetTextAngleAtIndex Method](#)
[IGTol::GetTextAtIndex Method](#)
[IGTol::GetTextCount Method](#)
[IGTol::GetTextFont Method](#)
[IGTol::GetTextHeightAtIndex Method](#)
[IGTol::GetTextInvertAtIndex Method](#)
[IGTol::GetTextPoint Method](#)
[IGTol::GetTextRefPositionAtIndex Method](#)
[IGTol::IGetTextPoint Method](#)

SOLIDWORKS API Help

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IGetTriangleAtIndex Method (IGTol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGTol Interface](#) : IGetTriangleAtIndex Method (IGTol)

Gets the triangle at the specified index.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetTriangleAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IGTol  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetTriangleAtIndex(Index)
```

C#

```
System.double IGetTriangleAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetTriangleAtIndex(  
    & System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the piece of text

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*vertexPt1[3]*, *vertexPt2[3]*, *vertexPt3[3]*, *isFilled*, *lineType*]

where:

vertexPt1[3] = First XYZ vertex point

vertexPt2[3] = Second XYZ vertex point

vertexPt3[3] = Third XYZ vertex point

isFilled = Boolean returned as a double and is True if the triangle is filled, false if not

lineType = Line type (see [swLineTypes_e](#))

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetTriangleAtIndex Method](#)
[IGtol::GetTriangleCount Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetWitnessLine Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IGetWitnessLine Method (IGtol)

Gets the extension line that connects the leader of this geometric tolerance with the entity that is being dimensioned.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetWitnessLine() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Double
```

```
value = instance.IGetWitnessLine()
```

C#

```
System.double IGetWitnessLine()
```

C++/CLI

```
System.double IGetWitnessLine();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to array of six doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

As you drag a geometric tolerance leader off an entity, SOLIDWORKS dynamically builds a witness or extension line that connects the leader to the entity.

The return value is the following array of six doubles:

[startPt[3], endPt[3]]

where:

- startPt[3] = x,y,z start point coordinates of the witness line
- endPt[3] = x,y,z end point coordinates of the witness line

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetWitnessLine Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBelowFrameTextAt Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : InsertBelowFrameTextAt Method (IGtol)

Inserts the specified text at the specified line index in the below frame of this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBelowFrameTextAt( _  
    ByVal Index As System.Integer, _  
    ByVal Text As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim Text As System.String  
Dim value As System.Boolean  
  
value = instance.InsertBelowFrameTextAt(Index, Text)
```

C#

```
System.bool InsertBelowFrameTextAt(  
    System.int Index,  
    System.string Text  
)
```

C++/CLI

```
System.bool InsertBelowFrameTextAt(  
& System.int Index,  
& System.String^ Text  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

1-based index where to insert the line of Text in the below frame

Text

Text to insert

Return Value

True if text line successfully inserted, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::InsertBelowFrameTextAt](#).

Example

See **Set Text in Datum Tags and GTols** examples in [IGTol](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::DeleteBelowFrameTextAt Method \(\)](#)
[IGtol::GetBelowFrameTextAt Method \(\)](#)
[IGtol::GetBelowFrameTextLineCount Method \(\)](#)
[IGtol::SetBelowFrameTextAt Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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IsAttached Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : IsAttached Method (IGtol)

Gets whether this GTol is attached.

.NET Syntax

Visual Basic (Declaration)

```
Function IsAttached() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Boolean  
  
value = instance.IsAttached()
```

C#

```
System.bool IsAttached()
```

C++/CLI

```
System.bool IsAttached();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if GTol is attached with a leaderline, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::IsAttached](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::GetLeaderCount Method](#)
[IGtol::GetLeaderInfo Method](#)
[IGtol::GetLeaderSide Method](#)
[IGtol::HasExtraLeader Method](#)
[IGtol::IGetLeaderAtIndex2 Method](#)
[IGtol::IGetLeaderInfo Method](#)
[IGtol::IGetLineAtIndex Method](#)

SOLIDWORKS API Help

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ISetTextFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : ISetTextFormat Method (IGtol)

Obsolete. Superseded by [IAnnotation::ISetTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ISetTextFormat( _  
    ByVal TextFormatType As System.Integer, _  
    ByVal TextFormat As TextFormat _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim TextFormatType As System.Integer  
Dim TextFormat As TextFormat  
Dim value As System.Boolean  
  
value = instance.ISetTextFormat(TextFormatType, TextFormat)
```

C#

```
System.bool ISetTextFormat(  
    System.int TextFormatType,  
    TextFormat TextFormat  
)
```

C++/CLI

```
System.bool ISetTextFormat(  
& System.int TextFormatType,  
& TextFormat^ TextFormat  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TextFormatType

TextFormat

Visual Basic for Applications (VBA) Syntax

See [Gtol::ISetTextFormat](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SeparateRequirement Property (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SeparateRequirement Property (IGtol)

Gets or sets whether a SEP REQ label is present with the Gtol symbol.

.NET Syntax

Visual Basic (Declaration)

```
Property SeparateRequirement As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

instance.SeparateRequirement = value

value = instance.SeparateRequirement
```

C#

```
System.bool SeparateRequirement {get; set;}
```

C++/CLI

```
property System.bool SeparateRequirement {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if a SEP REQT label is present, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SeparateRequirement](#).

Example

See the [IGtolFrame](#) examples.

Remarks

The Separate Requirement label is present below the Gtol frame.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetAllAroundThisSide Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetAllAroundThisSide Method (IGtol)

Sets whether this GTol uses an All Around This Side leader.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetAllAroundThisSide( _
    ByVal AllAroundTS As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim AllAroundTS As System.Boolean

instance.SetAllAroundThisSide(AllAroundTS)
```

C#

```
void SetAllAroundThisSide(  
    System.bool AllAroundTS  
)
```

C++/CLI

```
void SetAllAroundThisSide(  
& System.bool AllAroundTS  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AllAroundTS

True to use an All Around This Side leader, false to not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetAllAroundThisSide](#).

Example

'VBA Preconditions:

'Open a drawing.

```
Dim swApp As SldWorks  
Dim Part As ModelDoc2  
Dim myGtol As Gtol  
Dim myAnno As Annotation  
Dim boolstatus As Boolean  
Dim longstatus As Long  
Option Explicit  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set Part = swApp.ActiveDoc  
  
    Set myGtol = Part.InsertGtol()  
    If Not myGtol Is Nothing Then  
        myGtol.SetFrameSymbols2 1, "<IGTOL-SPROF>", False, "", False, "", "", "", ""  
        myGtol.SetFrameValues 1, ".031265", "", "", "", ""  
        myGtol.SetFrameSymbols2 2, "", False, "", False, "", "", "", ""  
        myGtol.SetFrameValues 2, "", "", "", "", ""  
        myGtol.SetPTZHeight "", False  
        myGtol.SetCompositeFrame False  
        myGtol.SetText 4, ""  
        myGtol.SetBetweenTwoPoints False, "", ""  
        myGtol.SetAllAroundThisSide True  
        Set myAnno = myGtol.GetAnnotation()  
        If Not myAnno Is Nothing Then  
            boolstatus = myAnno.SetPosition(0.801796955941255, 0.800162656875834, 0)  
            longstatus = myAnno.SetLeader3(swLeaderStyle_e.swBENT, 0, True, False, False, False)  
        End If  
    End If  
    Part.WindowRedraw  
  
End Sub
```

Remarks

This property is valid only for bent, perpendicular, and multi-jog leaders. To create bent, perpendicular, and multi-jog leaders, use [Annotation::SetLeader3](#).

Use:

- [IGtol::IsAttached](#) to determine whether this symbol is currently using a leader.
- [IGtol::HasExtraLeader](#) to determine whether this symbol is using a bent leader.

- [IGtol::GetLeaderSide](#) to determine where the leader is attached to the symbol.
- [IGtol::SetLeader](#) to set the characteristics of the leader on this symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetAllAroundThisSide Method \(\)](#)
[IGtol::SetAllOverThisSide Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0